



UNIVERSITI MALAYSIA SARAWAK

Faculty of Computer Science and Information Technology

Assignment/Report Cover Sheet

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Assignment Title:	Project Proposal		Lecturer:	Nurfauza Binti Jali	
Project Title: Ultimate Athletics Gym Management System					

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The Proposal of Ultimate Athletics Gym Management System (UAGMS)

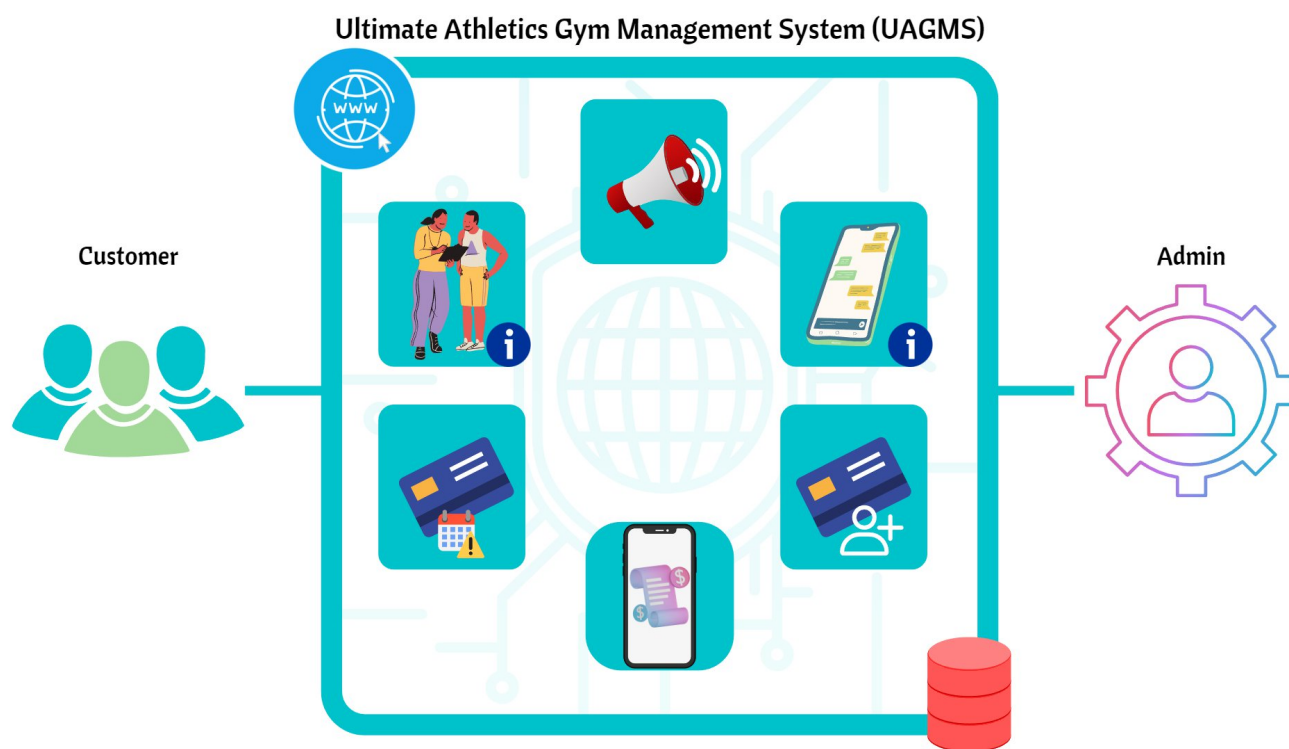


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Abstract

The official website is essential in today's technological and digital era for giving the customer the information they require and to ensure the validity of any kind of business for the company. For the administrative officer or operational manager to keep track of any specific data, a database system is necessary. Ads monetization and social media platforms like Facebook, Tiktok, Twitter and Instagram are used by our client, Ultimate Athletics Gym and Fitness Centre, to inform customers about their location and program offerings. Additionally, this business company does not use a database system of any kind for the membership application. The goal of this project is to address membership management, keep track of the membership progress and session in Eek Online Program and Eek Personal Program, and the company's legitimacy as a business. The development of a web application with a database system for Ultimate Athletics Gym and Fitness Centre has been agreed upon by our team, Castor, and our client, Ultimate Athletics Gym and Fitness Centre. This solution was discussed with the client and our team as possible solutions. Our team will develop and build a web application with a database system based on our client's requirements and specifications. We conducted interviews with the company's founder and operational manager in order to comprehend and learn more about our client's business. The client's requirements are crucially captured during the interviews session for the proposed solution, a web application with a database system. We were able to gather all the information required for the creation of the web application during the interviews that were conducted. With the help of this data, we were able to develop a quick plan for creating the suggested web application with a database system for our client. Throughout the project's development, our team hopes to be able to devote our full time and effort to creating this web application with a database system for our client, Ultimate Athletics Gym and Fitness Centre. We also hope that when the project is finished, our client's problems will be solved using the solution proposed by our team.

1. Introduction

Today's business issues demand dynamic business responses. In order to demonstrate the legitimacy of the business and to provide information to the customer, every company needs an official website in today's modern and digital era. A database system is also necessary for any business that needs to manage specific customer data. For information about their location and training programs, our client Ultimate Athletics Gym and Fitness Centre has been using ads monetization and social media sites like Facebook, Tiktok, Twitter and Instagram. Other than that, our client does not have any official website. Also, our client does not have any kind of database system to keep track of the membership application and information primarily on detecting membership expiry date. Other than that, our client also has a hard time keeping track of customer progress and sessions left to any particular customer.

Our team put forth the idea of developing a web application with a database system called the Ultimate Athletics Gym Management System to address the problems that our client is experiencing. Since December 9th, 2013, our client has been in charge of this gym and fitness centre. In August 2014, the gym and fitness centre was officially opened which is still open today. It is situated in Sarawak's Kota Samarahan. Our team will create the Ultimate Athletics Gym Management System, which is based on the needs and requirements of the client.

1.1. Company Background

The owner and founder of Ultimate Athletics Gym and Fitness Centre, Mr Mohd Ikrima bin Yahya, has nine years of experience in the health and fitness field. He launched this company on December 9th, 2013, and it has been operating since August 2014 officially. He and his other two closest friends came up with the concept (co-founder). In addition to being best friends, they also want to grow their partnership as business owners. They started the business with their own savings, not with a loan from a bank or any other institution. As a result, many people visited the gym and fitness centre in 2014. Due to this, the founder decides to purchase an additional room for fitness training and classes like Zumba and High-Intensity Interval Training (HIIT). The founder of this health and fitness company opened it out of a passion for business and fitness and a desire to assist others in improving their quality of life through a healthy diet and weight loss. He argued that prevention is preferable to treatment. This idea inspires him to encourage others to lead healthier lifestyles. Due to its advantageous location, this gym and fitness centre is situated in Kota Samarahan. This is due to the large number of diverse individuals who come here only temporarily, including university students, IPG teachers, and many others. Thus, it offers the business a wide range of opportunities and a promising future.

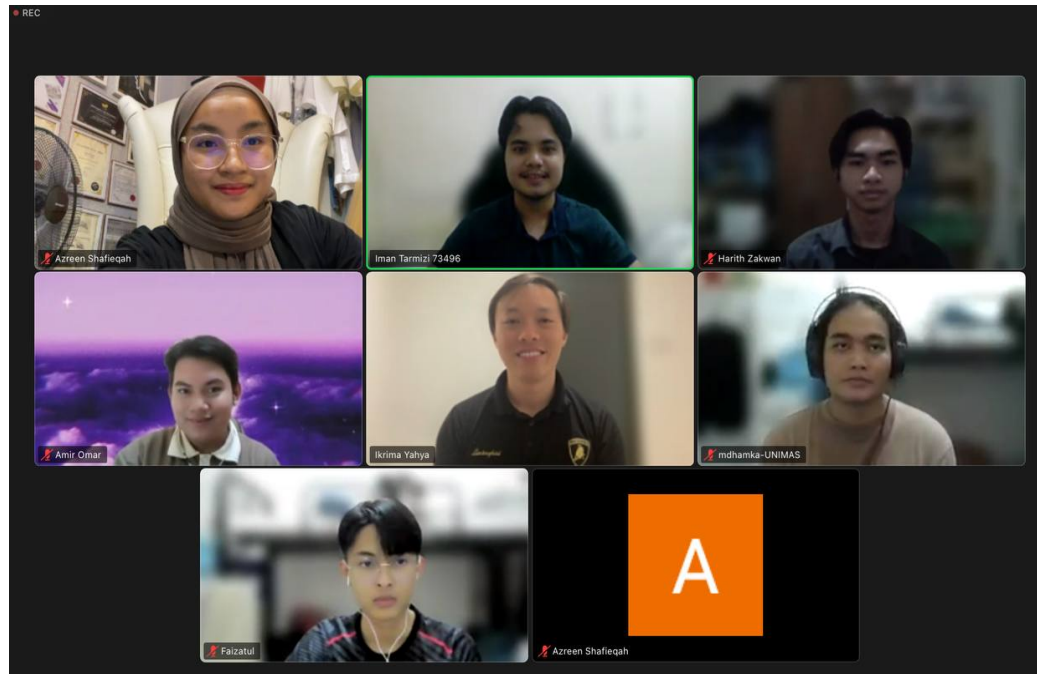


Figure 1: Interview Session with Founder in Zoom Meeting



Figure 2: Ultimate Athletics Gym and Fitness Centre logo

1.2. Team Background

The Ultimate Athletics Gym Management System project's driving force is made up of six people. The members name are Mohd Amir Alam bin Rahim Omar, Mohd Hamka Izzuddin Bin Mohd Yahya, Azreen Shafieqah Binti Asri, Harith Zakwan Bin Zakaria, Faizatul Fitri Bin Boestamam, and lastly Iman Tarmizi Rosalina. We gave this group the name "Castor" because it consists of six people, just like Castor, a prime ornament of the Gemini constellation and one of the brightest stars in the sky, is actually a system of stars with six unique members. The formation of this group began on October 23, 2022. All of the team members involved have been assigned roles and tasks based on their respective fields by Mohd Amir Alam Bin Rahim Omar. Each team member handled their assigned task as directed by the team leader while playing their part in the overall scheme of things. Mohd Ikrima Bin Yahya founded Castor with the intention of creating the Ultimate Athletics Gym Management System for the organisation known as the Ultimate Athletics Gym and Fitness Center. Under the direction of Madam Fauza Binti Jali, this project is for the TME3413 Software Engineering Laboratory course. The upcoming Ultimate Athletics Gym Management System will help the Ultimate Athletics Gym and Fitness Center increase their business legitimacy, provide information for their programs, services, facilities, company's background, coach information, provide e-invoice for membership and program, and to track down progress, amount of session, and membership applications, especially with regard to membership expiration date in line with the digital era and modern technology. We will collaborate with Ultimate Athletics Gym and Fitness Center on October 26, 2022. Having Ultimate Athletics Gym and Fitness Centre as a client on this project is truly a blessing for our team, Castor. In order to fulfil the requirements and standards established by this health and fitness company, we will therefore put all of our time and effort into doing so.



Figure 3: Team Member of Castor



Figure 4: Castor Official Logo

2. Problem statement

Since the country was proclaimed to be transitioning from a pandemic to an endemic, a range of people from all ages and backgrounds have been spotted hitting the gym, particularly at Ultimate Athletics Gym and Fitness Center . It is encouraging that everyone has begun to take an interest in their fitness and health, as seen by the large number of customers who visit the gym each day, including four of our group members. Throughout the workout, we saw various flaws and issues with the management of the gym, namely regarding customer records for memberships, and special programs.

This issue piqued our curiosity in speaking with the gym owner, so we set up two interviews with Mr. Qisthee, the operational manager and the head coach of Ultimate Athletics Gym and Fitness Center, in person, and Mr. Ikrima, the founder, through online video meeting, to learn more about their gym management. According to the interview, there are only three staff members which is the founder, the operational manager who also serves as a full-time coach, and an assistant coach. Additionally, there is no identification or badge to distinguish between customers who want walk-in service, membership, or their own customised program as they only imitate their client visually. Additionally, they lack a manual record that allows them to track their client membership and personal program history. We also found out that the gym management heavily depend on their social media and online platforms such as Facebook, Instagram, Twitter, for advertising and marketing to promote the gym program and package. They also used WhatsApp frequently to engage with the client, especially the one who applied for a special personal and online program as they would share everything that their program could offer such as weight loss and muscle building and development training and schedule. In essence, the issue with their data management system is that they do not have a database to store their data in and instead rely only on the WhatsApp platform to maintain and store customer data. If a client only applied for a membership, however, it would be difficult for them to determine when their membership would expire unless the staff had a manual logbook in which to record their information. According to what they told us, they only verbally registered member clients and collect payments in cash or online.

Therefore, we proposed a solution which is a management system that would solve their problem in managing their client program and membership registration and record and to create another platform to help them promote their program to the potential new client.

3. Project Objectives

Goal 1:

3.1 To study, analyse, and choose a workable solution for the client management issue at Ultimate Athletics Gym and Fitness Center.

Objectives:

- i. Study and comprehend the issue the client is having with managing their customer data through an interview.
- ii. The data and information gathered during interview sessions will be analysed, examined, and evaluated to find a solution to the problem.
- iii. Propose a software system as a solution that meets the client's needs.

Goal 2:

3.2 To design and implement a system based on a client's need and requirement.

Objectives:

- i. Create a requirement for the proposed software system based on the needs of the client.
- ii. Design the system's architecture, features, and functions based on the previously specified requirements.
- iii. Implement the system design.

Goal 3:

3.3 To test and evaluate the system that has been developed whether the system functions well and fulfills the system requirement based on client needs.

Objectives:

- i. Test the proposed system to see if it works properly and meets the needs of the client.

- ii. Evaluate the system's quality and functioning to enhance and correct the system's weaknesses and satisfy the client's requirements before the system will be hand over to the client.

Goal 4:

3.4 To maintain the system once it is fully operated by the clients in their gym if the system requires updates and repairs.

Objectives:

- i. Update the system to ensure that the customers may continue to use it without interruption.
- ii. Repair the system to guarantee that client operations were not disrupted by system failure.

4. Project Perspective

The Ultimate Athletic Gym Web Application is a web-based system designed to replace the obsolete and rudimentary Ultimate Athletic Gym system to satisfy and cater to the Ultimate Athletic Gym requirements. This system would be used to handle customer details, membership registration, and records, as well as to serve as a platform in addition to social media to help promote the program to potential new customers. The creation of this web application for Ultimate Athletics Gym and Fitness Center will make the setting and ambiance there easier to operate on a regular basis. Because of the development of this web application system, customers can now access and explore Ultimate Athletic Gym's membership and specialised personal programs with their special facilities and corresponding coach. Furthermore, this web application assists Ultimate Athletic Gym and its clients in developing a stronger relationship in terms of connectivity, since Ultimate Athletic Gym is following the trend of having an official e-commerce website for their company.

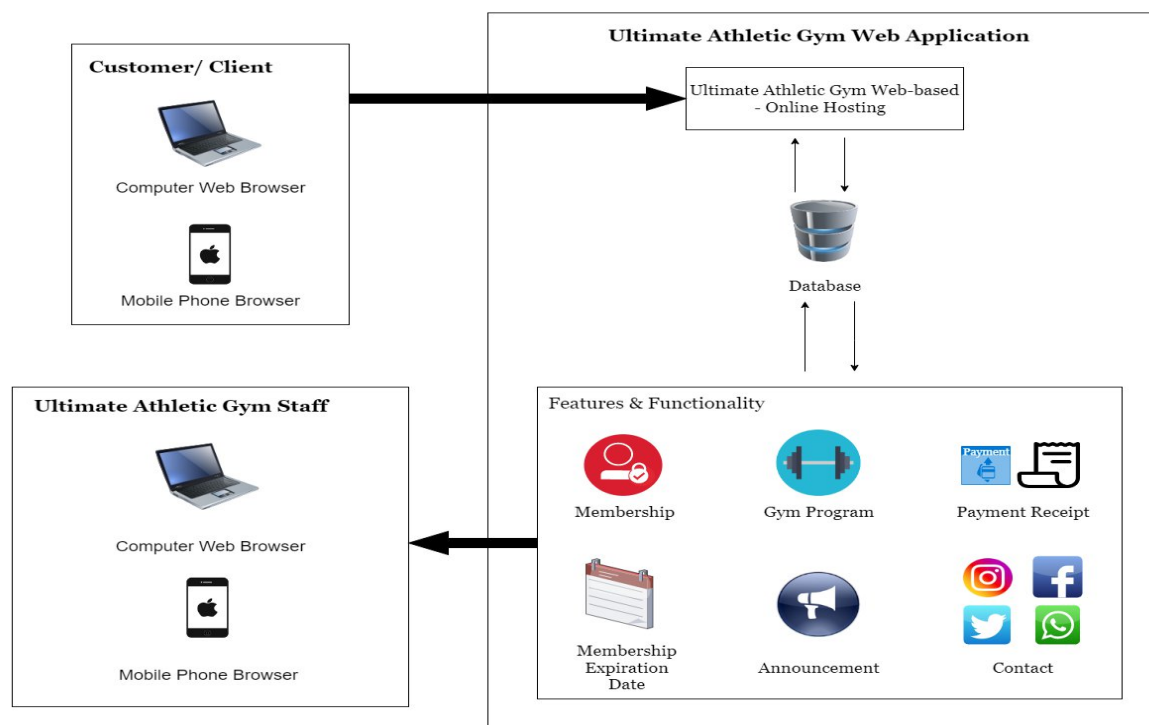


Figure 5: Ultimate Athletic Gym System Architecture

According to Figure 5 above, which depicts the system architecture of the web application system to be built, Ultimate Athletic Gym staffs and customers can access the web-based system via devices such as PCs, laptops, tablets, or even smartphones, as long as they have a built-in web browser and a stable internet access, but each user would have a various version of functionality as each user has different requirements in Ultimate Athletic Gym Web Application. Membership registration, customer payment, customer personal tracker, gym special program, gym services and facilities, company background and Ultimate Athletic Gym contact information are all included in the Ultimate Athletic Gym Web Application customer view while for the administrator/staff, it would have an advance functionality compare to the customer as they can perform an addition, reading, updation, and deletion which allows the administrators to manage customers data and record details. Administrators can add, view, edit, and delete programs detail, services and facilities, and membership prices as well as company information, from the database. Administrators may also view the membership expiry date and the history of e-invoices while updating the progress, session, and routine trackers.

Ultimate Athletic Gym Web Application web-based system provides the functionality for customers and administrator user as per below:

a) Customers:

- i. To view the Ultimate Athletic Gym dashboard, which includes their program, company background and contact information, as well as their gym services and amenities.
- ii. To view and update their profile.
- iii. To view their progress session and tracker in details.
- iv. To view their payment which is perform outside of the system by paying cash, online banking, or QR Pay and to generate an e-invoice after they have uploaded their receipt.
- v. To view membership details such as start date and expiry date.

b) Administrators:

- i. To view the Ultimate Athletic Gym dashboard of administrator with the features on the Ultimate Athletic Gym home page, including the membership application, special programs, customer information, and a page for updating services and facilities, programs detail, membership prices, and company information.
- ii. To manage customers data and record details. Administrators can add, view, edit, and delete programs detail, services and facilities, and membership prices as well as company information, from the database. Administrators may also view the membership expiry date and the

5. Project Scope

The goal of the web-based Ultimate Athletics Gym Management System is to make life easier for both the staff at Ultimate Athletics Gym and Fitness Centre and the customers. Clients can be anyone, as the service is accessible via the Internet. This website application will detail the various programmes and services provided by Ultimate Athletics Gym and Fitness Centre, as well as the fees. Ultimate Athletics Gym and Fitness Centre's background and an overview of the business will be presented to users and customers in the web app. The web app will also provide links to Ultimate Athletics Gym and Fitness Centre's social media profiles so that users may learn more about the gym.

We plan to implement main system functionalities both customer and Ultimate Athletics Gym's staff. For customer, they can view informations in the website such as reasons to choose this gym, view program available in the gym, view facilities offered by the gym, view services, view Ultimate Athletics Gym's background and contact, view tracker which are the customer can view session tracker and progress tracker, view own membership status and perform payment. For the client part, they can edit all the informations in the website, add new program, update membership details, add new register member, view payment history and update all the member's progression.

5.1. System Environment

The Table 2 below shows the system environment of Ultimate Athletics Gym Management System. The software, hardware and system requirements to satisfy the needs of the user for both user's end (Ultimate Athletics Gym and Fitness Centre's customers) and the client's end (Ultimate Athletics Gym and Fitness Centre's staff).

Environment	Software	Hardware	System Requirements
Development	Operating System - Windows 10	Personal Computer (Desktop / Laptop / Notebook) Mobile Devices (Smartphones / Tablets / Ipad)	Stable Internet Connection Location Access (GPS - Global Positioning System / LBS - Location-Based Service)
	Operating System - macOS Mojave		
	Source Code Editor - Visual Studio Code		
	Web Browser - Google Chrome		
	Web Hosting - 000webhost.com		
	Local hosting - XAMPP Server		
Delivered	Web Browser - Google Chrome		

Table 1: System Environment for the Ultimate Athletics Gym Management System

The Ultimate Athletics Gym Management System was created using the following development languages:

- Hypertext Markup Language, version 5

- Cascading Style Sheet, version 3
- ES6 JavaScript
- Hypertext Preprocessor in PHP
- MySQL

5.2. System Functionality

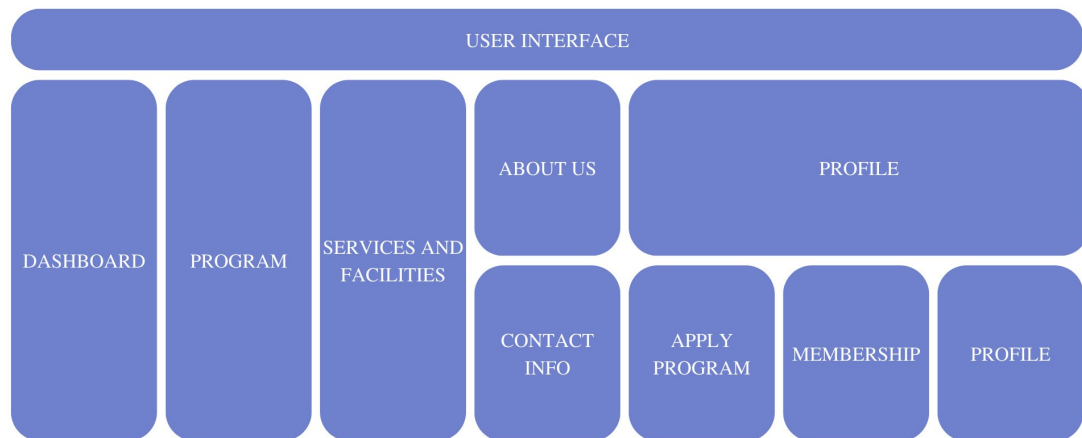


Figure 6.1: System Module For Customers

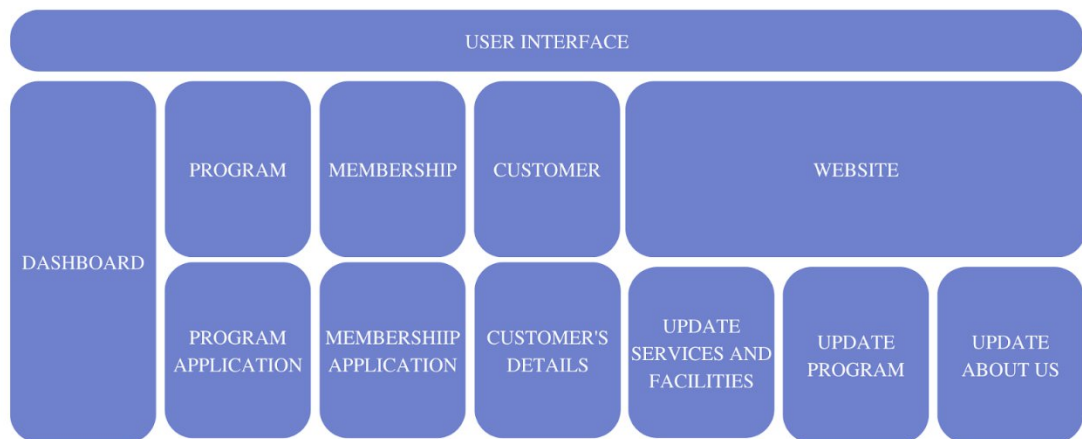


Figure 6.2: System Module For Admins

Module	Functions	Description
User Interface	The user should be allowed to interact with and use the online application	Displaying the system's features and functions in an easy-to-use way makes it easier for user to use
Dashboard	The user should be allowed to view home page showing the main program and reasons why to choose Ultimate Athletics Gym and Fitness Centre	• Video banner with slogan and button join now. • Simple description what are the specials about Ultimate Athletics Gym and Fitness Centre compared to others gym
Program	The user should be allowed to view program organized by Ultimate Athletics Gym and Fitness Centre	User will view the difference between program offered by Ultimate Athletics Gym and Fitness Centre with the button join now to make user easy to join the program
Services	The user should be allowed to view services offered by Ultimate Athletics Gym and Fitness Centre	• User will view more details of the difference between prices and the details that offered by the Ultimate Athletics Gym and Fitness Centre
Facilities	The user should be allowed to view facilities offered by Ultimate Athletics Gym and Fitness Centre	• User will be able to view pictures facilities and equipment in the Ultimate Athletics Gym and Fitness Centre
About us	To display the Ultimate Athletics Gym and Fitness Centre's information and coaches profile	• Users will be able to view the company's background, the coaches's background and their experience

Contact Us		To display the Ultimate Athletics Gym and Fitness Centre's contact information and social media	Users will be able to view contact information and location. Hyperlink to the company's social media sites such as Instagram, TikTok, Facebook page and Whatsapp number
Profile	Tracker	The sign up users should be allowed to view personal tracker	User can view their progress session and progress tracker in details
	Membership	The sign up users should be allowed to view membership	Users can view membership detail such as start date and expired date
	Payment	The sign up users should be allowed to perform payment	- User can choose to pay the program via transfer bank account, QR Pay or pay cash at the cashier - User need to upload receipt after payment - User will receive invoice after payment
	Profile	The users should be allowed to view own profile	Users can view and edit user's information such as email, password, full name, address, phone number. username and gender

Table 2: System Functionality for customer in Ultimate Athletics Gym Management System

Module		Functions	Description
User Interface		The user should be allowed to interact with and use the online application	Displaying the system's features and functions in an easy-to-use way makes it easier for user to use
Dashboard		The user should be allowed to view home page showing the membership details of Ultimate Athletics Gym and Fitness Centre	User can view table of membership details such as membership ID, invoice, name, date register and expiry date
Update	Program Update	The user should be allowed to update program show in the customer's website	User will edit program show in the customer's website such as program's name, program's image and description
	Progression Track	The user should be allowed to update progress session and progress tracker	User will add, view, and edit progress session and progress tracker each member
	Update Membership	The user should be allowed to update membership	User can renew membership for expired membership
	Update Info	The user should be allowed to update info	Users can update information in the customer's website such as why us, services, programmes, about us. and facilities.
View Progress	Session	The user should be allowed to view all member's progress session	- User can view member's progress such as session progression - Users can search which customer to view the details.

	Progress	The user should be allowed to view all member's progress tracker	• User can view member's progress such as and progress tracker • Users can search which customer to view the details.
Add Program		The user should be allowed to add new program	User can add new program such as program's name, image, and description
Register Member		The user should be allowed to add new register	User can register new member and need to fill information such as name, mobile, email and gender
Payment History		The user should be allowed to view payment history	User can view payment history such as member ID, invoice, payment method, date and file upload by customer

Table 3: System Functionality for admin's in Ultimate Athletics Gym Management System

6. Software Methodology

According to Stellman and Greene (2014), agile is a set of methodologies that can help a team think and work more effectively as well as assist in making better decisions. When it comes to the methodologies, they all share common traits to which all of them consist of practices that are streamlined and optimised that enable ease in adopting it (Stellman & Greene, 2014). Therefore, our team has decided on Scrum as our software methodology for this project due to its simplicity and high-performing nature.

The general gist of Scrum methodology is that it consists of a Product Owner, Scrum Master and Scrum Team (Stellman & Greene, 2014). The Product Owner works with the team for the maintenance and prioritisation of a Product Backlog that consists of features and requirements that are in need to be built (Stellman & Greene, 2014). Stellman and Greene (2014) states that the product is built through timeboxed iterations called sprints and at the start of each sprint, Sprint Planning is done by the team to determine which features that are included in the backlog will be built and this leads to a Sprint Backlog in which the team works throughout the sprint to build said features. A short meeting called Daily Scrum will be held by the team to update each other on the progress the team have made collectively as well as discussing any problems that may have arisen to which the Scrum Master will ensure the project keeps rolling by dealing with said problems together with the team (Stellman & Greene, 2014). At the end of the sprint, a Sprint Review will commence in which a working product will be demonstrated to the Product Owner then the team holds a Sprint Retrospective to reflect on any potential improvements that can be considered and implemented in future sprints (Stellman & Greene, 2014).

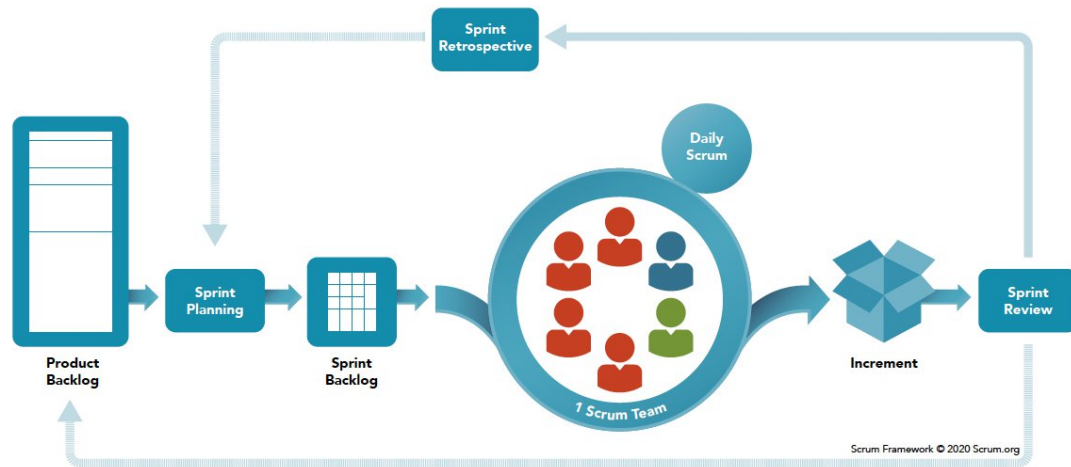


Figure 7: Framework of Scrum Methodology

As previously stated, our team has decided on the Scrum methodology because of its simplicity and high performing nature. Furthermore, Scrum methodology can complete project deliverables quickly and efficiently, favouring this project as a whole since the project duration is relatively short and the purpose of the project is to deliver a high performing product to our client. In addition, the sprint basis of Scrum methodology also serves as a great medium for our team to efficiently fractionate substantial and complex tasks into miniscule and simpler manageable tasks while keeping the quality high as well as taking the involvement of our client's conduct into account to ensure their satisfaction with the product can be reached by the end of each sprint and eventually the project.

7. Task Allocation & Contribution

Team Member	Roles, Task and Contributions
Mohd Amir Alam Bin Rahim Omar	1. Project Manager ● Coordinating the production release and roll-outs. ● Monitor the project's progress. ● Present the project's progress and results. 2. Design Manager ● Research and present effective design. ● Test the system design performance.
Mohammad Hamka Izzuddin Bin Mohamad Yahya	1. Planning Manager ● Planned and scheduled group meeting, meeting with clients, proposal, report, SDS, SRS, and Student Information Sheet completion for submission ● Supervise and ensure the progress made by the team is in line with the schedule. 2. Support Manager ● Lead the customer support department. ● Supervise the operations in customer support. ● Coordinate technical support for customers throughout the project. ● Assist and support the project manager and other members of the group in completing and progressing the project.
Azreen Shafieqah Binti Asri	1. Customer Interface Manager

	● Provide interface communication with customers from a technical standpoint. ● Ensure the interface is user friendly and well built. 2. Design Manager ● Define the design strategies. ● Create engagement model design.
Harith Zakwan Bin Zakaria	1. Implementation Manager ● Oversee the implementation of the product. ● Develop the system based on the client's request. 2. Customer Interface Manager ● Design the interface based on the requirement specification. ● Ensure optimum performance in handling multiple interfaces.
Faizatul Fitri Bin Boestamam	1. Test Manager ● Create test processes and strategies to analyse the working product. ● Provide improvement advice during development to reduce defects in product ● Review the system and search for any potential issues. 2. Support Manager ● Provide support in term of technical and psychological to other team member throughout the project ● Respond to any problems that occur during the

	project.
Iman Tarmizi Rosalina	3. Quality Manager ● Set the standards for product's quality. ● Inspect, test, and evaluate the product to ensure the product adheres to the quality standards. ● Monitor the performance of product and recommend improvement that can be implement. 4. Process Manager ● Evaluate and monitor business processes. ● Ensure efficiency in business processes. ● Create and implement changes in business processes.

Table 4: Task Allocation and Contribution

8. Team Goals

Upon further discussion with each and every team member, the 4 team goals that have been set are as follows:

1. To ensure that every given requirement that is requested by our client is fulfilled to guarantee their satisfaction.
2. To deliver the final product of this project on time at the end of the project's duration.
3. To maintain the quality of the product of high quality in each and every iteration throughout the project's duration.
4. To work together as a team diligently and to constantly support each other throughout the project's duration.

9. Expected Result / Deliverables

9.1. Working System

The solution for the working system that we will build for Ultimate Athletics Gym and Fitness Centre will be our expected result in completing this web application project. The Ultimate Athletics Gym Management System (UAGMS) will be implemented into the web application and is for both client and customer usage.

One of the features in the system for the customer is being able to display the gym's programs alongside the price, services and facilities, coaches and company background. After they choose which program or membership, they will prompt to make payment based on the price. There are 3 payment options which are payment using QR, paying in cash or bank transfer. Once the customer completes the payment successfully, customer needs to upload proof of payment or receipt in the web application. The client side will receive a notification that a customer has paid for their programs. The client will accept their membership or program application and update the customer application status. Then, an e-invoice will be sent to the customer's email. The website application also offers recommendation/announcement features such as events, promotions, or new programs to the customer. A signed-up customer will be able to use a progression track feature for personal

use or progression details from coaches for those who join the gym's program. A signed-up customer also will be able to view their membership information and update their profile.

On the other hand, for the client side (Ultimate Athletics Gym and Fitness Centre), the key feature is to be able to view the membership for tracking purposes such as expiration date. Any customer that successfully paid for the membership, the client will be able to register their customer as a member or renew the customer's membership. Another primary feature for the client is to update and add programs and information related to the gym on the customer's website interface so it is always kept up to date. In addition, the client will be offered a feature such as viewing and updating customer health details (session progression) who join the gym personal program will be used to track their progression according to the session they had. The progression tracker of gym members also can be viewed by the client. Moreover, the client can view the payment history between the client and the customer for business ledger purposes.

Our team expects this website application will help the Ultimate Athletics Gym and Fitness Centre to gain more customers and expand their influence on the Internet as we create the official website to increase their business legitimacy. Lastly, the web application is expected to be user-friendly, and well-designed to ease both customers and especially the client to manage their current situation as well in the future.

9.2. System Artefacts/Documentation

Artefacts / Documentation	Description
1. Project proposal	A documentation that defines the project, including things such as project requirements, objectives and goals. In other words, it is a document that proposes a solution in a project to the stakeholders, such as the client of the project and the lecturer.
2. Meeting minutes	Meeting minutes are a written record of the discussions and decisions made during a meeting conducted throughout the project making.
3. System's user manual	A document that shows to the user the steps of utilising and operating the system. This document will be the reference for both client side and software's tester.
4. Software Requirement Specification (SRS)	Is a document that completely describes what the software will do and how it will function. It also describes the functionality as well as non-functionality required by the product to meet the needs of the client.
5. System Design Specification (SDS)	The System Design Specification (SDS) is a comprehensive document containing all of the information required to develop the system. The process of defining the architecture, components, modules, interfaces, and data for a system in order to meet specified requirements is known as systems design. The application of systems theory

	to product development could be viewed as systems design.
6. TSPi forms	A tool to gather, analyse, and report the data for the project. It is also used to assign team roles and set up the goal of the project.

Table 5: System Artefacts

10. Project planning

No.	Task	Duration	Start	End
1.0	Planning	15 Days	25/10/2022	08/11/2022
1.1	Task distribution among members	1 Day	25/10/2022	25/10/2022
1.2	Gather Information From The Client	3 Days	26/10/2022	28/10/2022
1.3	Analyse Requirements	5 Days	29/10/2022	02/11/2022
1.4	Documentation of Project Proposal	5 Days	30/10/2022	03/11/2022
1.5	Review of the Project Proposal	2 Days	02/11/2022	03/11/2022
1.6	Submission of the Project Proposal	1 Day	04/11/2022	04/11/2022
1.7	Presentation of Project Proposal	1 Day	07/11/2022	07/11/2022
1.8	Sprint Planning Meeting	1 Day	08/11/2022	08/11/2022

2.0	First Iteration (Phase 1)	43 Days	09/11/2022	21/12/2022
2.1	Prototype System Development	31 Days	09/11/2022	09/12/2022
2.2	Features and Design Testing	3 Days	09/12/2022	11/12/2022
2.3	Initial Report preparations	7 Days	12/12/2022	18/12/2022
2.4	Initial Report Submission (WEEK 10)	1 Day	19/12/2022	19/12/2022
2.5	Initial Report Presentation (WEEK 10)			
2.6	Sprint Review Meeting	1 Day	21/12/2022	21/12/2022
3.0	Second Iteration (Phase 2)	20 Days	22/12/2022	06/01/2022
3.1	Modification and maintaining the system	5 Days	22/12/2022	26/12/2022
3.2	Testing the whole system	2 Days	27/12/2022	28/12/2022
3.3	Client Validation	2 Days	29/12/2022	30/12/2022

3.4	Minor Changes	3 Days	31/12/2022	02/01/2023
3.5	Final Test Run	2 Days	03/01/2022	04/01/2023
3.6	Closure Of Project	1 Day	04/01/2023	04/01/2023
3.7	Documentation of Project	3 Days	04/01/2023	06/01/2023
3.8	Submission of Final Report (WEEK 13)	1 Day	06/01/2023	06/01/2023
3.9	Preparation of Presentation	3 Days	06/01/2023	08/01/2023
	Final Presentation (WEEK 14)	1 Day	09/01/2023	09/01/2023
	System Deliver to the Client	1 Day	10/01/2023	10/01/2023

Table 6: Project Schedule

- Gantt Chart

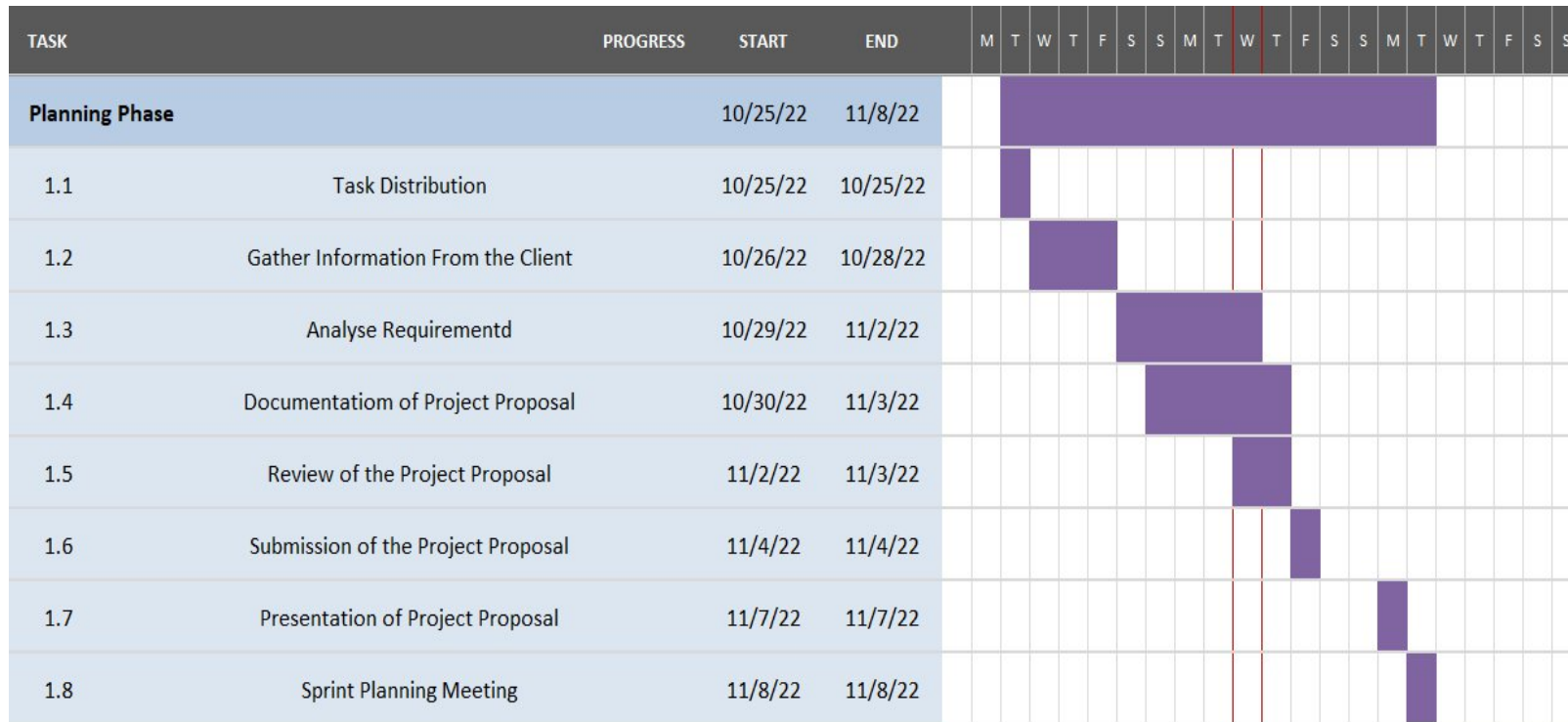


Figure 8: Planning Phase Gantt Chart

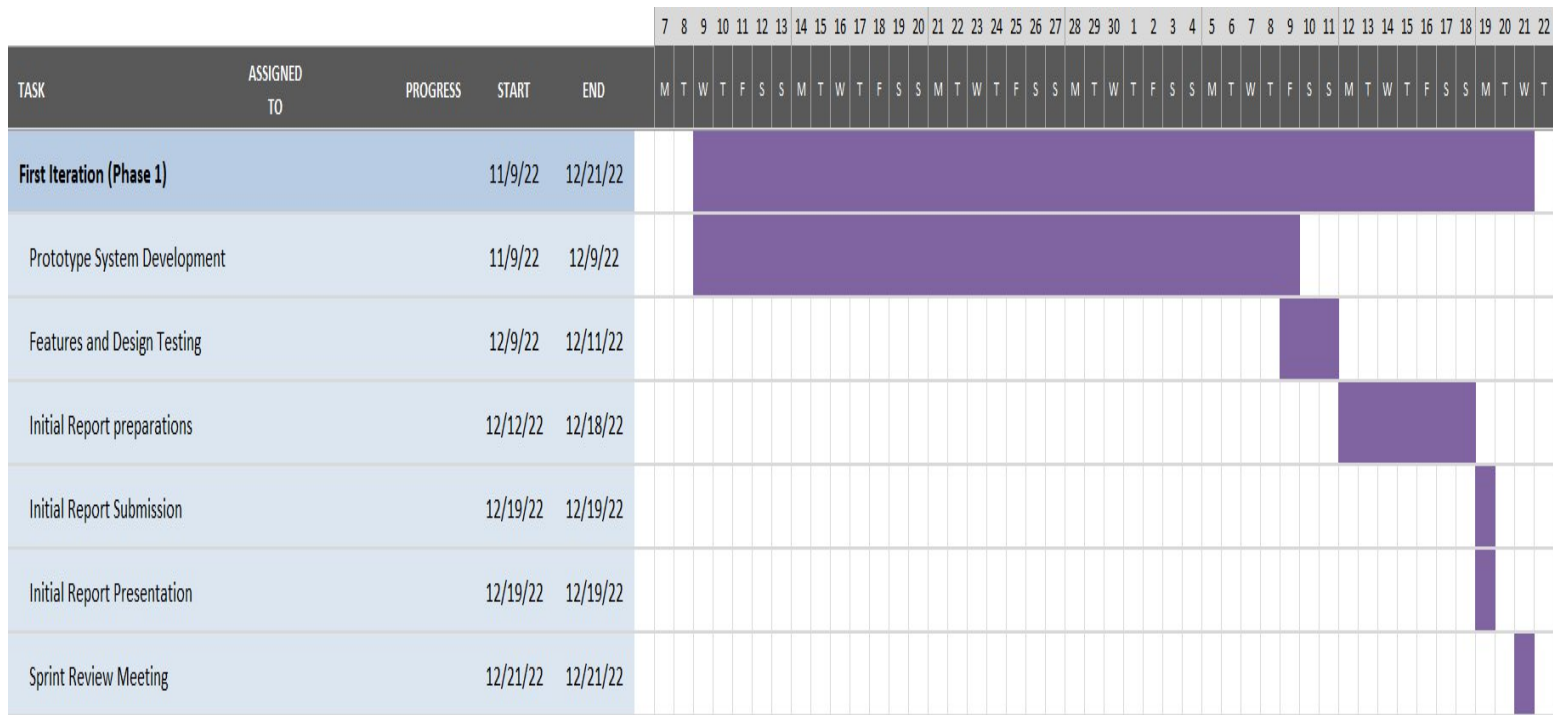


Figure 9: First Iteration (Phase 1) Gantt Chart

References

Stellman, A. & Greene, J. (2014). *Learning agile - Understanding scrum, xp, lean and kanban*.
O'Reilly Media.

Appendix

A. Interview with Operational Manager

1. How many staff and trainers do you have?

-We have 3 staff and our trainer which is:

- a) Mohd Ikrima bin Yahya - Founder and Admin
- b) Mohd Qisthee bin Yahya - Operational Manager and Gym Trainer
- c) Mohd Waquiuddin bin Yahya - Assistant Coach

2. Is there an organisation in this gym?

-Yes, there is. The three staff consists of the gym owner who manages the finances. The other person, who is me, is a full-time trainer who manages client progress. The third is an assistant trainer. The organisation can be shown as below:

- a) Mohd Ikrima bin Yahya - Founder and Admin
- b) Mohd Qisthee bin Yahya - Operational Manager and Gym Trainer (full-time trainer)
- c) Mohd Waquiuddin bin Yahya - Assistant Coach (part-time trainer)

3. What makes this gym special from other gyms?

-Our gym has a very spacious space compared to other gyms. For example, although this place has gym equipment, the space is very spacious. In addition, what is special about this gym program, which is a program to lose weight, we have achieved a 100% success rate with the right method.

4. How many customers follow the program at the gym?

-At least 10 people and above follow the program.

5. What are the privileges of a customer if they become a member?

-If the customer often goes to the gym, we recommend that them to take a membership instead of walk in because the price is more cheaper and worth it. Membership can come to the gym every day and at any time by paying only RM90. Meanwhile, walk in is RM15 for a day.

6. How do I renew an expired membership?

-We only use the whatsapp platform to collect customer data. For example, if a customer makes a membership today, we will record it on WhatsApp. After that we will see on WhatsApp whether the membership has expired or not.

7. Since this gym uses the whatsapp platform, Has you ever been a problem with whatsapp like you lose all data?

-So far, this has never happened before. This is because all three of our staff store the data in WhatsApp group. We feel it is impossible if all staffs lose customer information.

8. If a customer joins a personal coach program, are they allowed to choose a trainer?

-Not allowed. As I told you earlier, we only have 3 staff members. I am the only full-time trainer. Others rarely go to the gym. If I can't train customers in large numbers, I need an assistant trainer.

9. What programs are implemented at this gym?

-There are two programs. The first, (EOP). This program is for weight loss program but online. 100% online without having to come to the gym. We use the Whatsapp Group platform. The second, (EPP). Same with weigh lost program. Customers need to come to the gym. We as trainers will help them. At least they need to take 3 sessions.

10. Does this gym open a vacancy to work at this gym?

-For now not yet. But we plan to find a clerk to manage admin matters. Previously we were looking for a trainer to work with us. But unfortunately the trainer does not have good criteria to become a trainer. For example, he knows about working out but he is not good at teaching others. Maybe in the future, if our customers get more and more, we will consider adding trainers. And we have to filter carefully.

11. How do you promote this gym?

-We use Facebook, Instagram and TikTok. Sometimes we also use ads in Facebook.

12. How much is the service here?

-Walk in - RM15/day

-Membership - RM90/month

-Coach Program- RM30/session

13. How is the customer program schedule at this gym?

-The schedule is not fixed because we follow the hours or days when the customer is free.

14. How do you identify if this customer has a membership or not?

-We only imitate visually.

15. How many customers can you handle at the same time?

-Usually only 4 people. But if the customer prefers private or group, we can fulfill their request.

B. Interview with the founder

1. Who is the founder of this company?

-Mohd Ikrima bin Yahya.

2. Who got the idea of this business?

-Mohd Ikrima and his two bestfriends.

3. What inspired the founder to choose this business?

-Mohd Ikrima and his two bestfriends have the same business-minded. They also want to success together in a business. Hence, they inspired indirectly with each other besides reading and watching motivational books and videos.

4. When did the company start this business?

-Started on 9th December 2013 and officially open in August 2014.

5. Why did the company choose this business?

-Passion in business and fitness. Also, they want to help people to have a better healthy lifestyle.

6. Why in Kota Samarahan?

-Strategic location due to most people lives here temporarily and people always changing. For example, after a person finish learning or teaching in Kota Samarahan, he or she will go back to the hometown.

7. Where did the owner learned fitness knowledge?

-Research and reading books, following a lot of fitness trainer worldwide and asking their guide. Then, the knowledge gained will be applied to himself. Besides, the owner have Zumba Instructor Certification from Brunei Darussalam.

Student Information Sheet

TSP Student Information Sheet: Form Info

Name: Mohd Amir Alam bin Rahim Omar

Instructor: Nurfaeza bt Jali

Date : 29 October 2022

Number of College Credits:

Major: Software Engineering

Expected Graduation Date: October 2024

Briefly describe your relevant experience and interests:

I have studied back-end and front-end coding during my first and second year studies in Unimas. I have experienced with C, C++, Java, JavaScript, HTML and PHP. Full stack developer is what I want to achieve. I want to train myself to be able to both back-end and front-end.

Briefly describe your work on other team projects:

I have worked for both back-end and front-end programming during my previous projects. I usually will do the planning and designing all the features needed in the system. I also did some testing to see whether the system is fully functional or not. If not, I will debug the code.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

I did involved in many Co-curricular activities during my Secondary and University. For example, I became student prefect during my high school years. During my pre-university, I became a Project Manager for some projects such as Historical drama project and Bazaar Asasi. During my degree, I am one of the JPK (Jawatankuasa Perwakilan Kolej) and JPF (Jawatankuasa Perwakilan Fakulti) known as PERTEKMA.

State your team preferences, if any:

I want to work with people who knows their responsibility about doing work as a team and knows how to give suggestion and take advice from other members.

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.

Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
0800-0900		TMF3113		TMF3113			
0900-1000	TME3424			TMF3113			
1000-1100				TME3413			
1100-1200			KMU1013				
1200-1300		TME3033	KMU1013				
1300-1400		TME3033	KMU1013				
1400-1500		TME3033	GAMELAN				
1500-1600			GAMELAN				
1600-1700	TME3413		GAMELAN				
1700-1800	TME3413	TME3424		TMF3012			
1800-1900		TME3424		TMF3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:

Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5
Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5

TSP Student Information Sheet: Form Info

Name: Mohammad Hamka Izzuddin Bin Mohamad Yahya **Instructor:** Madam Nurfaeza Jali

Date: 25/10/2022

Number of College Credits: -

Major: Software Engineering

Expected Graduation Date: 2024

Briefly describe your relevant experience and interests:

From what I have learned in my degree, I have experience in Front End and Back End Development. My interest in theoretical computer science such as algorithm, data structures, mathematics, statistics, Object-Oriented Programming, Boolean algebra, and Logic Gates and in applied computer science such as C language, Java, C++, Web Development language like HTML, CSS, and JavaScript, and database management system.

Briefly describe your work on other team projects:

Throughout my time in pre-university and university, I participated in many team projects, but I believe that my time as a student of software engineering was particularly difficult and demanding, especially the Web-Based Development, Java, and Database projects. I participate in such projects as a front-end and back-end developer, both of which need for a strong command of the relevant programming language.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

I have held a variety of positions during my time as a student in secondary school, pre-university, and university, including high council positions like president, vice president, secretary, and treasurer as well as positions on the academic and student development executive committee during my pre-university years as a member of the JPPPU and the activity and volunteerism executive committee during my university years as a member of PERTEKMA. Additionally, I typically take on the roles of project manager, planning manager, and support manager throughout my time at pre-university and university.

State your team preferences, if any:

I am seeking for someone that is dedicated, enthusiastic, intelligent, encouraging, trustworthy, and highly determined. Essentially, I want a group or team that will have complete faith and confidence in my ability, knowledge, and skills to complete my task/work, and who will constantly encourage one another to be better and improve as a team, as well as who will not be afraid to point out my mistakes and areas of weakness so that I can strengthen my teamwork skills

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.							
Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
0800-0900		TMF 3113		TMF 3113			
0900-1000	TME 3423	TMF 3113					
1000-1100				TME 3413			
1100-1200			KMU 1013				
1300-1400		TME 3033	KMU 1013				
1400-1500		TME 3033	KMU 1013				
1500-1600		TME 3033					
1600-1700	TME 3413						
1700-1800	TME 3413			TMF 3012			
1800-1900				TMF 3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:					
Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5
Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5

TSP Student Information Sheet: Form Info

Name: Azreen Shafieqah binti Asri
Date : 25 October 2022
Major: Software Engineering

Instructor: Madam Nurfaiza Jali
Number of College Credits: -
Expected Graduation Date: 2024

Briefly describe your relevant experience and interests:

I have experience in design user interface during my previous group project. I interested in designing.

Briefly describe your work on other team projects:

In other team project such as course Web Based System Development, I am one of design interface. I love to play with colour and shape to make the website more attractive and user-friendly. I also a person who want to make the website with no mistake interface.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

My previous assignment course English for Professional Communication, I become a leader. My job scope is to manage the assignment, distribute the task, and find solution to the problems. All the assignment submit before the due date.

State your team preferences, if any:

I am looking for a hardworking member and not easy to give up.

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.

Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
0800-0900		TMF3113		TMF3113			
0900-1000	TME3423	TMF3113					
1000-1100				TME3413			
1100-1200			KMU1013				
1200-1300		TME3033	KMU1013				
1300-1400		TME3033	KMU1013				
1400-1500		TME3033					
1500-1600							
1600-1700	TME3413						
1700-1800	TME3413	TME3423		TMF3012			
1800-1900		TME3424		TMF3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:

Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5
Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5

TSP Student Information Sheet: Form Info

Name: Harith Zakwan Bin Zakaria

Instructor: Nurfaeza Binti Jali

Date: 29 October 2022

Number of College Credits: -

Major: Software Engineering

Expected Graduation Date: October 2024

Briefly describe your relevant experience and interests:

My experience with computer science in general started when I was in secondary school and during my last 2 years in secondary school, I have learned the basics of programming and everything that is associated with it. Prior to pursuing my major, I have had experience with scripting languages such as PHP, JavaScript and CSS as well as the markup language, HTML as they were used in projects at the time but my previous semesters have given me an excellent exposure to programming languages such as C, C++ and Java. My interests are mostly in dealing with the inner workings (planning, implementation and quality testing) of a project.

Briefly describe your work on other team projects:

My work on other team projects throughout the semesters has been mostly anything that is backend related. I am usually in charge of building the framework from scratch as well as dealing with the implementation, quality and testing parts of the projects as I have prior experience in programming which allows me to apply my existing experience in them.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

The only leadership experience was in highschool when I was holding the position as a student prefect. There were a lot of student prefects ranging from Form 1 to Form 6 so we were divided into

groups of said students and I was the leader of my own group. I was in charge of distributing the tasks of my members, planning and scheduling as well as supervising them.

State your team preferences, if any:

I prefer a team that is diligent and a team where everyone's collective effort compliments each other's work in the bigger picture.

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.

Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
800-900		TMF3113		TMF3113			
900-1000	TMF3423			TMF3113			
1000-1100				TME3413			
1100-1200			KMU1013				
1200-1300		TME3033	KMU1013				
1300-1400		TME3033	KMU1013				
1400-1500		TME3033					
1500-1600							
1600-1700	TME3413						
1700-1800	TME3413	TMF3423		TMF3012			
1800-1900		TMF3423		TMF3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:

Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5
Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5

TSP Student Information Sheet: Form Info

Name: Faizatul Fitri Bin Boestamam

Instructor: Nurfaeza Binti Jali

Date :25 October 2022

Number of College Credits: -

Major: Software Engineering

Expected Graduation Date: 2024

Briefly describe your relevant experience and interests:

I have experience in the back-end and front-end development of a program during my years of studies at UNIMAS. From the perspective of coding work-related, designing the interface was much easier but I want to train myself more in developing the back end of a program/ website. My interests would be implementation, testing, and quality control in the project.

Briefly describe your work on other team projects:

During my studies in this major, I worked in the development team on other team projects to create the program or website. I usually handle the planning, implementation, quality control, and testing part of the project.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

I have held many management positions during my studies. The nearest example is I was the deputy of exco academic in the Student Representative Committee (JPP) during my foundation level in Unimas. I was in charge to deal with any academic problems that the students find during their studies.

State your team preferences, if any:

I prefer a team that motivates each other and try our best until the end.

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.

Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
800-900		TMF3113		TMF3113			
900-1000	TME3424			TMF3113			
1000-1100				TME3413			
1100-1200			KMU1013				
1200-1300		TME3033	KMU1013				
1300-1400		TME3033	KMU1013				
1400-1500		TME3033					
1500-1600							
1600-1700	TME3413						
1700-1800	TME3413	TME3424		TMF3012			
1800-1900		TME3424		TMF3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:

Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5
Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5

TSP Student Information Sheet: Form Info

Name: Iman Tarmizi Rosalina

Instructor: Madam Nurfaeza Jali

Date: 31st October 2022

Number of College Credits: -

Major: Software Engineering

Expected Graduation Date: 2024

Briefly describe your relevant experience and interests:

In my previous group project, I gained experience designing graphical user interfaces and databases. Additionally, I have experience using PHP, including PHP myAdmin, JavaScript, CSS, and HTML. I am therefore qualified to create both client-side and server-side software. Being a full stack developer is something I'm very interested in.

Briefly describe your work on other team projects:

I was a GUI and server software designer on a previous team project for the Web Based System Development course. I fulfilled my responsibility by creating the customer's payment method and payment information and connecting all the payment details in database.

Briefly describe any leadership or management positions you have held (at work or in clubs/organisation):

I took charge of my MathX group project while I was still a foundation student at UNIMAS. My responsibilities included overseeing the project's proposal and work flow, setting up team meetings with the lecturer and team members, assigning tasks to team members, and managing survey data for the project.

State your team preferences, if any:

I'm looking for someone who is dedicated, reliable, creative, outgoing, and good at solving problems. A person who never gives up in any situation is also desirable.

List your class schedule and other times when you have scheduled activities such as work ,clubs, sports teams, etc.

Time	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
0800-0900		TMF3113		TMF3113			
0900-1000	TME3423	TMF3113					
1000-1100				TME3413			
1100-1200			KMU1013				
1200-1300		TME3033	KMU1013				
1300-1400		TME3033	KMU1013				
1400-1500		TME3033					
1500-1600							
1600-1700	TME3413						
1700-1800	TME3413	TME3423		TMF3012			
1800-1900		TME3423		TMF3012			

Rank from 1 (least) to 5 (most) your preferences for serving in the following team roles:

Team Leader	1	2	3	4	5
Development Manager	1	2	3	4	5
Planning Manager	1	2	3	4	5

Quality/Process Manager	1	2	3	4	5
Support Manager	1	2	3	4	5